

A1. [2 marks] Let $P_2(x) = -2 + 5(x - 3) + 3(x - 3)^2$ be the degree-2 Taylor polynomial of $y = f(x)$ centred at $x = 3$. Find the equation of the tangent line to $y = f(x)$ at the point $(3, f(3))$.

A2. [2 marks] Let $f(x)$ be a function with the following properties:

$$f(4) = 3, \quad f'(4) = -1, \quad f''(4) = 6, \quad f'''(4) = -5$$

Write the degree-3 Taylor polynomial of $f(x)$ centred at $x = 4$.

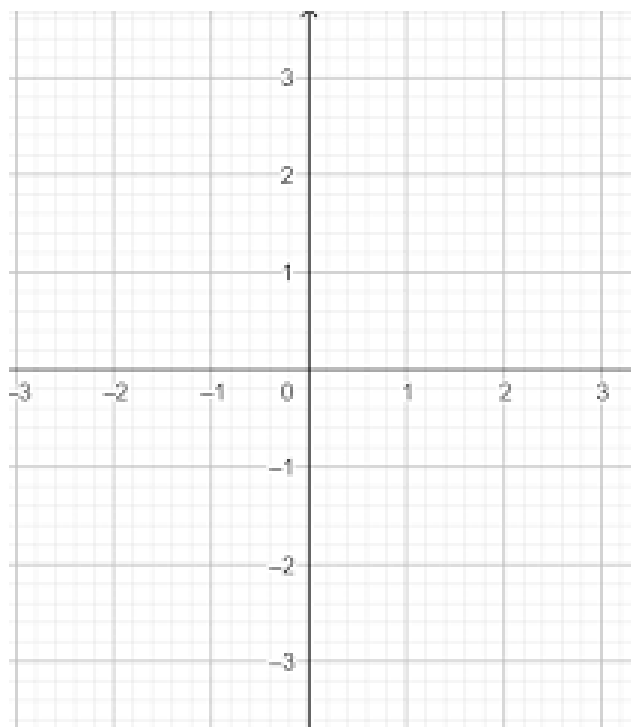
A1. [1 mark] What is the largest finite value of b for which the following integral is improper?

$$\int_b^1 \left(\frac{2 + \sin(x)}{(x+1)^2} \right) dx$$

A7. [2 marks] Consider the differential equation

$$\frac{dy}{dx} = y^2 - y - 6$$

The solutions of this differential equation are functions of x . **Sketch the graph(s) of the equilibrium solution(s) on the axes below. DO NOT SKETCH ANY OTHER SOLUTIONS.**



A8. [3 marks] Let $P(t)$ be the population of bacteria in a petri dish after t seconds. The rate of change of this population with respect to time is proportional to the number of bacteria in the petri dish. Write an example of a differential equation involving P that agrees with the previous sentence.

B1.(a) [1 marks] Does the following integral converge or diverge?
Circle the correct answer:

$$\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$$

B1.(b) [2 marks] Write an integral of the form $\int_1^{\infty} \frac{1}{x^p} dx$ that can be compared to $\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$ in order to explain why your solution to part (a) is correct.

B1.(c) [3 marks] Explain in 1-5 sentences how your answer from (b) can be compared to $\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$ to conclude your answer from (a).

B2.(a) [3 marks] Find the general solution of the following differential equation. Show your work.

$$\frac{dy}{dx} = 2y - xy$$

B2.(b) [3 marks] A particle's velocity $v(t)$ at t seconds is measured in millimetres per second, where

$$\frac{dv}{dt} = 2v - tv$$

For this question $t \geq 0$. The initial velocity of the particle is 6 mm/min. Use your solution from part (a) to find the velocity function for this particle. Show your work. You can refer to your answer from the previous question.

B3. Consider the differential equation

$$\frac{dy}{dx} = \frac{x}{2y}$$

where $y(2) = 6$.

In this question, we will approximate a solution for y .

B3.(a) [1 mark] Calculate $y'(2)$

B3.(b) [2 marks] Calculate $y''(2)$

B3.(c) [3 marks] Use your answers to parts (a) and (b) of this question to calculate a Taylor polynomial of degree 2 for the solution of the differential equation y , centered at $x = 2$.

B3.(d) [1 mark] Use your approximation to write an expression that is equal to an estimated value of $y(2.25)$. You do not have to simplify this expression.